

ActInf Livestream #013.2: "Cybernetic Big Five Theory with the Free Energy Principle"

Source: [YouTube](#) **Playlist:** Active Inference Livestreams (Paper discussions)

Duration: 2:09:26 | **Uploaded:** Unknown **Transcript method:** auto_caption

all right hello everyone welcome to the active inference live stream it is active inference live stream 13.2 on january 12th 2021 today is going to be a really awesome discussion so thanks everyone who has come out to participate welcome to the active inference lab we are an experiment in online team communication learning and practice related to active inference you can find us at our website twitter gmail youtube or our keybase public team and public username and this is a recorded in an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and as far as video etiquette for live stream goes mute if there's noise in the background raise your hand so we can hear from everybody who wants to speak and we'll use respectful speech behavior all the active streams in 2021 at least the regularly scheduled ones are on tuesdays from 7 to 9 pacific time so check out this [rb.gui](#) link to see the schedule today we are in 13.2 on january 12th with adam saffron and colin deyoung talking about this cool paper about cybernetics and personality and next week and beyond we're going to be heading into different sets of papers so check out this spreadsheet and see if you're interested in participating um douglas i'm retired from information technology all right i just had to mute douglas so yeah please don't play a stream while you're on a live stream this is a cognitio 2021 and this is a conference in 2021 that is being organized by several of our colleagues uh by mao and bayern s who has appeared on the discussion before welcome blue and so this looks like a really cool conference that they are all organizing so check out this conference if you're curious and hopefully we'll be able to have updates on this conference later on so in the intro in warm-up session we introduce ourselves we can just give a short introduction or warm-up and then pass it to somebody who hasn't spoken yet so hi everyone i'm daniel i'm in california and i'll pass it first to blue hi everyone i am blue knight i'm an independent research consultant based out of new mexico oh and i'll pass to steven hello i'm stephen i'm based in toronto i'm doing a practice-based phd at the southern school institute of applied psychology and i will pass it over

to sasha hi i'm sasha i'm based out of davis and i'm a neuroscience graduate student and i will pass it to colin hi there i'm colin de young i'm a professor in the psychology department at the university of minnesota and i will pass it on to adam hi i'm adam saffron i'm a postdoctoral researcher out of bloomington indiana good to be with you ooh go to alex yeah hi everyone i'm alex i'm in moscow russia and i'm researcher in systems management school and i texted to scott thank you hello everyone my name is scott david i'm at the university of washington applied physics lab where i'm the director of the information risk research initiative and i'll pass it to marco hi thank you my name is marco arcolin i'm from holland i graduated in colorado science and i'm just very very interested in active infants i'm not affiliated at the moment but i'd like to be here um is there anyone left okay i had to kick out somebody who's doing feedback so please have it figured out and yes cool all right maybe not all right hello uh dave are you there okay he just got disconnected i am here hello hello uh okay well just go just proceed i'll figure it out fun great times thanks for being on so for the warm-up questions people can just raise their hand if they want to say something that's related or unrelated uh the same questions from last week will put up and take a little bit of a different go at them so what is something you are excited about today and what is something that you liked or remembered about the paper uh anyone can go for it i'd be curious to either of the authors just looking back on it it's not so long ago but we are in a new year so with the fresh lens of the numerology of the year and just maybe these discussions how are you looking back at what you had done and seeing it in a new or a different way or any other participants who want to add something that they thought of go ahead oh i i might as well go ahead then i was just going to say that i don't know that for me maybe even more so than for adam this particular chapter was sort of right at the boundary of the things that i've been thinking about and working with adam has really pushed me into um you know making contact with this whole topic of active inference and the kind of you know fristonian world um and then it's also made me think more about about modeling um because i've generally been fairly skeptical about the value of trying to model whole agents that are analogous to actual human beings uh for learning things about psychology just because uh because the brain and people are so complex you know if you if you have a model it has to have a sufficient degree of verisimilitude in resembling the thing that you're modeling before you're likely to actually learn new things just from studying the model rather than learning about you know the limitations of the model in one way or another and your skillet modeling but i guess one of the things that writing this chapter with adam made me think about was mars different levels of analysis and the way that if you get a good

enough uh resemblance at the level of uh of function or computation you might be able to learn things even if uh you don't have varisimilitude at the level of implementation uh or algorithm right we don't necessarily have to say that the brain is doing the exact kind of mathematical computations that that you know fep demands and in the same way that we would do them to model it but as long as we're capturing something that truly resembles the functional that a cybernetic system like a human being needs to carry out we might actually be able to learn new things about the way that people work cool steven and then anyone else who wants to raise their hand but just maybe we could hear from everybody in the warm-up so steven and then adam and then anyone else i'm really excited by the fact that this is a you know taps into a very widely used tool in psychology so and then links this back to these new potentials of active inference which you know we've talked a lot about maybe in more theoretical philosophy philosophical ways but this really is showing um how it you know the twos can start to move into each other so i'm curious today to see more ways in which you know there's questions about um you know the potential for that going forward you know what what possibilities in the field of even organizational psychology which uses these traits as well it's this these these traits are kind of used beyond kind of work with um sort of therapeutic environment so um and you know even with the myers-briggs and other types of things how can a more dynamic or kind of embodied you know what's the future basically for applied psychology who adam then scott um so this actually working with colin on this um it really stretched me also was that the like limits of what i was used to thinking about and uh i think i never like really seriously engaged with personality science before this like i was always interested but then i think i was also like skeptical that they had any uh chance of carving nature at his joints and i was just like uh maybe and but then as we started exploring the correspondences between like the cybernetic view of personality and the other cybernetic perspective of active inference i started to become more compelled and then i'd say like things are always kind of moving around for me in terms of like uh perspectives but um i think recently some of the things i've been thinking about is like how far can we stretch or apply this these ideas of personality over uh what scales and what ranges of systems uh can we think of the personalities of societies the personalities of uh any system uh and so that's i think most recently been uh on my mind but i'm excited to be here because uh uh i love these live streams and uh last week was awesome cool agreed scott and then sasha than anyone else so a couple of things the the the active inference is totally new to me and it's really been revelatory as other people have said a couple of things i used to work in identity a lot still do identity kind

of systems federated identity kind of technical stuff not on the technical side what i realized is the attributes are carried in these markov blankets right so the you know the the access to attributes is through this the statistical views of the surface of other systems and so that's been really useful to me and then the other part i realized is i realize that a lot of folks um fristen and others are coming at this from the biology perspective of seeing gee how do these things emerge by themselves what i realize is as a lawyer all i've ever done is construct synthetic markov blankets because contracts are revealing a surface of a system in fact the system doesn't exist until the contract exists forget corporations if i want to put a supply chain in place i write a contract and then the organism of the supply chain exists as a result and so it's funny because in that idea of if it's too rarified for viewing human mind i get it and it's interesting because in the de-risking context where a lot of folks are getting together they try to make it simpler you know worry about bankruptcy worry about notice worry about these specific things boom boom that's all they care about so it's kind of like a synthetic organism constructed by the contract the markov blanket defines the organism it's like the opposite of the blind men and the elephant where they're trying to find the elephant here there is no elephant until there are other parameters so the the attributes define the system and it's a very bounded system very few attributes to it but it simplifies it and that's what a lot of regulatory market issues are about so that's been huge for me really realizing that we can synthetically construct these systems and base them on shared story shared narrative shared meaning contracts as a formal way of doing that but folkways is an informal way of doing that and i think that's absolutely a very rich avenue for for my work and i'm looking forward to working with others on it thanks cool thank you sasha and then anyone else on the warm-up yeah sasha then adam okay um yeah uh i'm just really kind of following up on what adam mentioned um thinking about other applications for um the big five theory and what other groups and organisms they might apply to and i think uh very relevantly for this group and of the kind of projects that we're working on is um personalities of a team um and uh thinking about different behaviors i'm i'm sure this isn't a novel concept um but i'd love to hear more about it from the authors of other possible applications um of these metrics for uh teams of varying sizes um and the other thing i've been thinking about since last week is this um if this paper aims to integrate these two concepts of active inference and um the big five theory um are there parts that are incongruent is there something that doesn't um that can't work together from both of these systems and is there something that active can glean or kind of take away from personality research in a way that you know hasn't really been tried before cool questions

thanks adam then blue uh somewhat flipping uh what i said before related to uh what sasha and scott just said um so like in also not just applying personality to other systems where you might not normally think of them in terms of um i'm also interested in like reversing that flow so it's like we how might we think of the personality of teams but the things we know about teams how might that influence how we think about persons and their personalities or uh contracts like what are analogies or models that could be derived by thinking of what you described or this like inactive contract like creating the mark of like the contract in the in the doing um to what extent can we have this as part of um person formation and personal evolution personality evolution um i don't know if that makes sense but that's another place i've been thinking is like this bi-directional flow of like taking things from persons putting them in terms of things we might not normally think of as persons taking things we might not normally think of as persons and then putting it back to persons and their personalities yeah it's kind of like finding the human in the non-human atom really interesting point so personifying and enlivening things that are perceived as inanimate and then finding the mechanism inside of living systems² so blue then colin so going back to what adam was saying not just now but earlier um and also just something that i liked about the paper so for me psychology has kind of always been like this soft science like you can't really do science on psychology but really like what that means is that it's a wicked problem right it just falls into the complexity realm like people are super super intricate complicated it's a convoluted like thing so this paper for me really um was awesome in the way that it kind of concretized or hardened up for for me like a quantitative way to um really look at personality interesting colin than anyone else and i'll put up the third question as well well first i'm really glad to hear that this might have convinced somebody that psychology wasn't just soft science uh it's it's definitely hard science in the sense that it's difficult um and that's largely about you know issues well it's two things i think it's issues with measurement um the the kinds of measurements that we want to do and can do you know they don't necessarily line up very well in terms of precision and so you know everyone has physics envy because you've got precision down to 10 decimal places on measurements but you know we're lucky if we have precision down to uh down to one decimal place or if it's even clear that our measurements are you know anything other than essentially heuristic you know if you have a if you have a scale and you're measuring a personality trait and essentially what you've got is just a measure of people's uh you know degree of a particular kind of behavior or experience relative to other people so the measurement is relative and it's not clear that the you know one point on the

scale actually corresponds to uh some true interval rather than just a kind of heuristic approximation so yeah there are a lot of really difficult challenges in psychology but nonetheless i mean in order to do science you just have to have some kind of standardized measurement and then be able to uh to replicate right to see if other people can get it i used to think that you had to have uh you know hypothesis testing too but then i realized you can really just do science if you do it exploratory work and then replicate it because once you've found something that then de facto becomes the hypothesis for the replication attempt so all you're really looking for are reliable patterns in reality fundamentally and i think we can do that to some extent in psychology i wanted to say something also about uh sasha's question about the you know whether there are incompatibilities between um fep and active inference and um general approaches to personality and my theoretical approach specifically um and i think one of the things that's really nice about active inference is that it's uh it's it's very low level in some sense it's this formalism that is extremely widely applicable um and the way that i see it is that it's essentially a formalism that captures the essence of all cybernetic systems um and so you know i haven't found anything that makes me say oh this isn't gonna work you know it's like there's a lot of detail about what people are that isn't captured in active inference but only because it's so low level but like there's nothing about active inference that isn't crucial to what people are right because you know that the process of uh you know pursuing goals via feedback that is just necessarily the core of what an agent is and so i guess my concern isn't that they're incompatibilities it is how to draw out the ai framework in such a way that it makes contact with a lot of the specifics that we know about humans function rather than the generalities about how all cybernetic systems function really interesting answer we'll go to dave and then markup hi my professor of cybernetics gordon pasque in the 1950s and 60s developed some very fine-grained measures of learning strategies which is how individuals approach a specific concrete task as they're learning it and learning styles which is a more systematic way that folks uh develop a fixed repertoire of approaches to a task that the extremes of the on that scale are serial learning and holistic learning i believe the the areas of his data where he uses uh entropy as an important measure for certainty and for tolerance for ambiguity should tie in with um the um active what the not not the active insurance the other the other major area that we're talking about uh free energy and i suspect that the learning styles should correlate with personality and that is an area that i believe he uh past did work on so that's something i need to dig into the literature on and compare it cool marco hi thanks um also thank you scott that was very well put um so in general i just really like a lot of things that

have been said um about active influence as a as a framework i i also echo the notion that the beauty is that it's so uh generally applicable that you can kind of elaborate it and that becomes a kind of game of if you take a bit further with a few assumptions uh when does it touch or overlap or align with other existing years or empirical data but what has been mentioned and something i find personally an intriguing possibility is um not necessarily active inference but if something like active influence would be true and it would indeed be the most approximate description of our neural dynamics then naturally follow that it also can become quite intuitive so just like colin said that you can elaborate until it kind of touches something else in terms of theory you could also maybe consider the idea that you can elaborate your understanding of active influence relative to personality or other psychological notions until start to touch your kind of social of people of yourself of others society and i think that's what i think someone in the beginning said i think i think was stephen about the future of applied psychology and i think this is really really crucial not just from a scientific perspective but also a social and humanist perspective because i mean i think most people agree that folk psychology isn't enough and i think also most people agree that most scientific psychology whether it's a neuro or clinical side isn't very easily implementable uh in terms of actual understanding yourself by first person so i think that's definitely the most exciting aspect i think this particular work is a very nice milestone with regards to that i mean i don't think anyone really understands itself that much better with uh some ratings on the big five but knowing the mechanisms i think is a very very beautiful lens to understand myself cheers thank you for sharing steven and then anyone else yeah so following on from what marco was saying though is like the the fact that this paper also takes in the affordances and and active influence like you say does this dynamics you you gives this generative potential to the big five which i know is kind of sort of seen to be there but you know the the challenges that they can end up being whether you like it or not prescriptive and kind of diagnostic to some extent and having a way to relate them opens up that phenomenological sense because there's a i think what i'm really interested is if people structure their metaphorical models of how they cognized in the world like how they whether they relate to time as time coming towards them or they're going towards time or whether they're um trying to um build a a more firm career or they're trying to get on in the world and travel and go somewhere so those kind of phenomenological ways of structuring thought they they seems to be more ways that you can then connect them as opposed to it being like completely disconnected or just too big a gap to bridge so this uh this mixture of the affordances the dynamics the blanket and the attention and the way that phenomenological um

sort of structuring could happen even at the individual level starts to become more plausible which is kind of cool awesome yeah a lot of cool ideas coming up one theme to return to is that by looking at the patterns and thinking about the frameworks that span different systems of different levels of analysis we can think about like the traits of organizations teams individuals organs cells so that's the multi-scale kind of complex systems approach because we set out to look for patterns across systems and then that led to all these interesting questions like would there be a big five for teams or how are the generative models of individual personality related to the generative models or the generative processes underlying group personalities and so pretty interesting things that we're coming to any other points in this intro before we just return to some of the later pages colin go ahead and then anyone else i'll just make one point in relation to this issue of uh you know team dynamics and personalities at other scales um that's something that there are different ways to approach it right so people have certainly psychologists have certainly studied uh what happens to teams that have different characteristics of the individual members uh and so you could think of how do the personalities of individual members of a team influence the success of the team um so for example people having people high in openness on the team tends to uh make it more effective in various ways and it seems to be because that's especially true for leaderless teams right so if you have a it's not necessarily true if there's a clearly established kind of leadership structure in the team but if you have leaderless teams then having people who are more sort of cognitively flexible enables them to kind of organize themselves better so but i think that what we're thinking about here is something a little bit different like is there a personality of a team as a whole that might be more than the sum of its parts um and you know they're you're probably they're probably going to be well you you need to do a sort of parallel kind of research in terms of trying to come up with a comprehensive just set of descriptors of the properties of teams and then seeing if they varied systematically in the same way that people did with individuals and leading to the big five but i guess one of my intuitions is that the highest level of the personality hierarchy stability and plasticity which are also the most replicable across different languages when people have done the kind of big five research so there's this very clear way in which uh what i think of as the two fundamental properties of all cybernetic systems you know at least ones that have to be able to function in an unpredictable and complex environment that is probably applicable at virtually any scale even for something that might not be well it's i think it's an interesting question whether something like a team or a society is actually a cybernetic system itself or whether it is an emergent dynamic out of a set

of cybernetic systems interacting with each other but regardless i think you can certainly describe the stability and plasticity of teams or even societies right you can describe them in terms of how much they focus on and value the lack of disruption of their functioning versus how much they value expansion and exploration and dynamic growth cool that reminds me so much of the ant research and do colonies have phenotypes or is it just the phenotypes of nest mates to selection shape colony traits or is it just shaping the nest mates and they're duking it out so all these multi-level systems we kind of can see these similarities so adam and then scott and then anyone else so in relation what collin just said um correct me if i'm wrong but i believe in terms of like cross-cultural like the robustness um like is it the stability and plasticity like this like that level of granularity actually is what is the most broadly applicable in terms of like showing the least variation like if you look at it in those two yes so in terms of the lexical research like starting from a specific uh language and extracting descriptors and then doing the same kind of you know ratings and factor analytic work the only truly universal solution that's been found is the two-factor solution and those factors very closely resemble stability and plasticity okay so yeah so that would support the idea like if we're trying to do like a multi-scale personality story or a quasi-anthropomorphizing of of uh of quasi agents or maybe we should just call them agents depending on our view that yes we will be most likely to find stability in plasticity but um you know because like you basically grounded the big five as part of cybernetic functioning and if any uh cis like from an active inference perspective if any like sustaining uh system is doing some kind either explicit or implicit goal pursuit we actually might find like even the big five itself like being surprisingly applicable although like as you're saying like it's an empirical question and so like so i guess the idea is we actually do the research program of this sort of derivation of factors and and with and then we should expect to find things might be uh shockingly like the big five and places where you might not expect them nice i didn't raise my hand but i just wanted to come in and say like i think the best analogy for that that's going on in science right now is uh over the last let's say 20 years people have really started to take seriously the idea of personality and other species now i mean whether personality is the right word there or we want some other one it doesn't really matter right you have individual differences in psychological functioning of other species and so people have been doing this attempting to come up with you know extensive sets of descriptors of other species and uh you know ratings of them or coding based on you know video or observation or whatever and then looking to see whether there are these similar patterns and most of the big five are identifiable uh or you know fairly close analogs in and

um the one that is the least commonly found is actually maybe sort of you know proof of concept in a way because conscientiousness is the dimension that least appears in other species something like conscientiousness and and in fact the only two species where something close to that has appeared as its own separate dimension is in bonobos and chimpanzees right and so you think like what really makes humans special well it's our ability to plan for the distant future and to prioritize goals that aren't uh me immediate or near immediate and to really have these elaborate goal hierarchies and so it might not be surprising that it's only our nearest evolutionary neighbors that have these kind identifiable patterns that are uh that are related to those abilities and that are separate from things just like you know how emotionally reactive you are very interesting scott and then i think adam and then marco so that this is great this um one of the things i thought was interesting is like you know what gets measured gets done and what gets measured gets seen and so it's kind of interesting because we're coming it's like a bio theory of everything kind of idea here and we're like wow this is really explanatory my recollection is that quantum mechanics was just a calculation convention originally but it kind of works and we kind of use it and even though we don't understand all of it it's kind of helpful that way so it's not so much is it true but is it useful and it's this is broadly useful and so it's interesting in living systems or we start to get at these questions we ask and when we ask questions from the i don't know i'd like someone to tell me what the big five is after i make this comment but the um the if they're the big five of consciousness then the um i always think of them as the accounting firms but that's just because i was a lawyer for so many years but um but the uh what are we it's like we're using the turing test in a way it's too much hubris about our form of intelligence right and so if we point our measurement instrument at a certain thing that's we're going to see if we point our measurement instrument at human cognition human consciousness our sense of human consciousness that's what we're going to see and so this is a great discussion there's no it's not a to me it's not a truth-finding thing but a usefulness of the paradigm kind of issue and if someone could tell me what the big five are just list them that would be good because i'm not familiar with the term as in parlance sure it's the big five dimensions that are empirically observed when you ask people personality trait surveys and their openness conscientiousness extroversion agreeableness neuroticism ocean and then one metaphor on the big five and uh it's kind of like the big five continents that's what we have in the world as we have it and so that is specifics that's related to history and that changes but the the principle is water and land and so maybe in a different system a different sphere with a different type of water level you still would have water and land but the

actual number of big dimensions might be empirically seen to be different so that's kind of an interesting thing um i think adam then marco then colin uh i don't know if this would be useful but perhaps in a bit actually might be good to have colin talk about like the specific mappings of the big five onto uh cybernetic function that he describes in his original paper uh that could be useful for people but it's interesting about um so conscientiously being less likely to generalize potentially across species or systems because uh it seems like that makes some sense to me and that um in terms of preserving uh goal hierarchy integrity as you're confronted with all these different like circumstances but you want to maintain this overall ex like broadly applicable gold hierarchy that's benefiting you overall um in order for that to be the case you have to have a system capable of generative modeling with or modeling with a certain degree of temporal depth and the policies it's pursuing and counterfactual richness such that it could actually get itself into trouble and get confused and go down like wrong paths and so you wouldn't expect this for all systems you would expect this you need a certain degree of sophistication even maybe sophisticated effective inference in order for you to really actually need something like conscientiousness so that one actually might not generalize as much um the and related to what scott just said um there seems to be this yeah interesting combination it's like realist versus instrumentalists or it's like you can think of like this personality is almost like uh like the big five is like an effective field theory over persons um where it's um a good approximate description um a a useful course screening but now the question i would have though is but is it like just completely so there's a sense in which it's modeler relative it's based on our interests but there's like another sentiment which it's like well um like what worked in evolution and like it's like you don't have to take the perspective of evolution but it's a we can all kind of converge on it's like quasi objective like it's one of many perspectives but it's a pretty darn good perspective or an important one to keep in mind and so in that sense the big five actually might be carving nature at its joints in a really powerful way and so i guess one more thing along those lines is um uh can be good for probably column to talk about this since you discovered them uh which is the aspects um and so one thing i'm wondering is uh so it seems like in a lot of studies when you do things at the level as a correct me if i'm wrong but it's kind of um so under so the big five has this hierarchical structure over them you'll have these two metatrades which we've been discussing of stability as shared variants of conscientiousness agreeableness and inverse neuroticism and plasticity a shared variance of extroversion and openness but then underneath the big five um column that you'll get if you like force if you'd like to have the water level uh go there's a down or up i'm i'm messing up

daniel's like analogy but you'll get um two and only two aspects underneath each of them and what's interesting is they oftentimes kind of read as being across purposes with each other it's like these are very different not across they're very different it's very different to be like industrious versus orderly like these are very different ways of being in the world or if like you're neurotic it's like very different to being like um withdrawing or being like more volatile and like reactive like um and so i guess one thing i'm wondering is like both in doing um active inferential modeling of persons and trying to suss out would it be better to actually in some ways focus on the aspects in terms of finding joints there but also maybe across other cybernetic systems like could there be more purpose uh purchase from the aspects who and i just flipped to the slide so just for anyone can just say go to this figure or go to this other thing so colin and then marco sorry you're muted thank you for the reminder um yeah so while there's a lot to uh think about in what's been said in the last couple of comments now um so in terms of this idea of uh of continence and water and land i do think that there's a way in which uh you know stability and plasticity are like the equivalent of of water and land so that pretty much uh any organism you're going to be able to identify you know if there's any kind of individual variability which of course there must be in order for evolution to proceed like so probably um you know if you take amoebas for example they're not going to have a lot of different personality dimensions but nonetheless you might find some that are a bit more of the same species that are a bit more exploratory and others that are a bit more uh more cautious let's say so you know even at this very simple level you may get this trade-off between uh do you do things that keep your functioning running uh without disruption or do you do things that lead to greater uh expanse and uh discovery of new resources so those probably do trade off i would hope that when we're within a species though that the the big five are not um like just the continents in other words they're not just a historical accident i think that they actually reflect variation in parameters that have evolved and that are present in uh in every you know intact organism of that species the issue though in terms of carving nature at its joints is just that the nature in this case is so complicated right so uh if you look at this is a good figure actually that uh daniel put up to make this point which is that um when we draw these kinds of hierarchical taxonomies of personality traits uh they are necessarily over simplifications especially at the levels below the big five because we know that personality does not have simple structure and what that means from a factor analytic perspective is that if you took the taxonomy uh exactly literally that's depicted there you would say well for example uh no trait that is underneath extraversion that's a subcomponent of extroversion can

be related to any trait that is a subcomponent of agreeableness right because there are no lines connecting them in that figure but in fact the pattern of association and covariation among more specific personality traits below the big five is considerably more complicated than that would suggest and so for example assertiveness is negatively correlated with politeness right so assertiveness is part of extraversion politeness is part of agreeableness those are correlated with each other however it happens that the enthusiastic or sociable component of extraversion is positively correlated with the other aspect of agreeableness which is compassion and caring about other people and so you have two different components of extraversion correlating in opposite directions with two different components of agreeableness and those end up canceling each other out and so at the first at the other level you see no correlation between agreeableness and uh essentially so this whole thing you know the structure it looks simple in the picture but uh you know it's a lot more complicated uh the the lower down you go and so i would have you know i it seems to me that these aspect level distinctions are important uh they have they're they are supported by some behavioral genetic research as well as some of my own research just at the phenotypic level um i think that they're probably the most important distinction within each of the big five in terms of discriminant validity in other words you often get things that are predicted differently at that level um so for example uh politeness and compassion are the two aspects of agreeableness and well it turns out that uh if you look at people's political orientation people who identify as more left-wing or liberal uh score higher in compassion people who identify as higher in uh conservatism uh more right-wing score higher in politeness right so conservatives are more polite liberals are more compassionate well what happens when you go up to agreeableness you see no correlation with political attitudes which people found surprising for a long time because it's you know issues around altruism and cooperation seem very relevant to political issues and policy well so it turns out that that's because something lower than the big five actually is associated with uh with political attitudes in opposite directions so i definitely think that these lower levels of the hierarchy can be useful um you know there's less the the evidence for the big five is remarkably extensive right whereas the evidence for that next level down that i've worked on is much less extensive nonetheless you know again i think there isn't just one way to carve this system up because the patterns of inter-relatedness are so complicated and i do think that these are useful uh useful sub-dimensions at any rate even if you could focus on others so i'll i'll leave that there cool thanks marco then scott thanks um so the last few comments were very very thoughtful um so so let me know if i ramble on for too long um so so i love metaphors so

if you don't mind i'll try to provide a complimentary one so instead of maybe landscapes um i have this thing with imagining the brain as one big dynamical system and so all those dynamical flows or patterns they need to of course be a pattern to order structured and adaptive sense so this kind of trick structure could maybe be seen as kind of a kind of grip kind of handle kind of heuristic um well of gravity to kind of heuristically bias particular dynamics and at least for me uh that would make sense because of these um big five factors um being quite quite segregated in terms of where i think they would be relevant and that gives me back to the matter of personality and and kind of anti-moralization because i agree that there's a big problem of our theory laden in the manner in which we approach psychology um and that also uh should exhibit itself in as mentions the observations when it comes to animals um but for example when it comes to conscientiousness as adam already very nice pointed out there's a matter of death and i think i think he did allude to it but another thing to emphasize is also um the kind of field or or a range of possible alternatives to to care about to someone who would be low on contentiousness or carefree would perhaps uh not simply be lacking goal-orientedness but be more um aware of in touch or have a bigger affinity with the notion of play of simply uh engaging in behavior that is reducing expected free energy yet um not necessarily defined in terms of a particular goal right so not all animals would have to deal with that right because the evolutionaries simply aren't trained or evolved to have uh uh awareness or sensitivity to a multitude of possibilities in what to do in environments and so the very emergence of a particular big five factor might be also related to the complexity or the kind of yeah the complexity of the imperatives that they may encounter in their life in their niches so one may be interesting example is like dogs or other social animals uh although in general i don't think it would be appropriate to project or impose uh our entrepreneurial notion of personality i do think at least dog owners can agree that dogs tend to have a personality that seems to develop or evolve and even seemingly be similar to their owners possibly because the imperatives of their environment is expanded by being in relation with human agents and so by the expansion of of possible niches possible situations um the relevance or the variation uh specific to what consensus is is a relevant factor uh becomes more apparent whereas uh previously that might not be the case if that makes sense um yeah i'll leave that that because i forgot all everything else i wanted to say no it's really interesting that it's uh big five is not the essence of an individual it's actually in a cybernetic framework making the individual's phenotype in relationship to the goals and the context of their niche so it actually moves away from some of the essentialist types of critiques that people might naively bring up against big

five we're looking at this as an empirical pattern with a generative underlying cybernetic reality so scott then adam than anyone else so fantastic conversation the couple of observations one personality is for a person and so it's kind of interesting when we take personality and impose it on other systems right i'm just playing linguistics there a little bit but in a way we what is the delta between personality and living formality i wonder so that might be in because are we assuming the human elements again it's that subjectivity question one of the things also it occurred to me that fristen it kind of allows for a notion of situated beyond situated cognition and so maybe those are overlapping maybe they're not i'm not sure if all um personality uh elements are seen as cognitive or if there's some things that are you know effective and hormonal and whatever other causation chains there are so many it's interesting that the um the behavioral elements we ascribe the personality um may or may not may not drive uh through all living systems i tend to define living systems and it's not anything reasoned it's just what i've done for the last 10 years and it's been effective for me is i define living systems as auto catalytic and entropy secreting so they can be in a variety of different forms and so i think of markets and corporations as living systems and so to me one of the questions that with with the definitions in the fristen notion we have kind of a fractal notion of uh things that are show scale independence i guess is one of the elements of fractal systems and so to what extent can the definition of using these personality traits at different levels help us understand the scale and dependent possibilities there so that's one issue then the last point these are kind of random but i think they're clustered around the idea of the model ascribing the last one is you know when we talk if i'm talking to daniel there's at least four identities involved there my sense of me his sense of me his sense of him and my sense of him so one of the challenges when you're talking to corporations markets maybe not dogs because they may have quality is that the other um the other entity doesn't have qualia if you're talking to a corporation it does not have quality the people in the corporation have qualia but but the corporation itself does not and so the um well that's obviously we can talk about that but my point is if they don't have qualia can they have empathy and if they don't have empathy maybe there are differences in kind up the fristen stack of fractal stack of living things and so maybe that is one way to start to distinguish between markets and corporations and again gets back to the animals issue do animals have qualia is the quality of their quality is slightly different than ours such that maybe there's a subset of personality that they'd show things like that um anyway just a couple of random thoughts i think but they again that definition of living beings i think is bound up in here and again not being to us not being too stubborn in our perspective

that the personality is something that would describe all the attributes that work up and down the biological stack thanks cool we got adam then stephen marco colin so go ahead adam hi um hmm okay there's a lot uh to try to write it in for me um i like the so what marco was saying with this um basically this this attractor network formation perspective the supposed metaphor or a really good model the he mentioned like niche construction and he mentioned um i i think this perspective is really important in terms of thinking of the dynamic character of the processes that might give rise to these attractors over the course of development and situations and with development potentially being understandable in an open-ended sense that never stops um and i guess along those lines i'm thinking of that causal process model that colin has in the original paper where it's going from genes and environment to you're having these cybernetic function and then traits and characteristic adaptations but then there's this feedback as and one way of understanding this feedback is as a kind of niche construction relationship where you're modifying your environment you're specializing yourself for these um as you're trying to like get along in the world as a goal seeking system um one more nice thing um with relation to what uh scott was saying i think that's actually uh yeah so i don't know also like while i'm interested in seeing like how far we can stretch personality i wouldn't want to overstate it either because there were major transitions in evolution and you know one of them would be um basically the ability to have certain types of generative modeling where there might be a qualia and quality of different varieties and then we might go beyond that not just qualia but things like access consciousness and if you like that distinction i do like being able to know your experience and report on it and things like being autonomic and being able to reflect on even your knowing um and so this would be very different cybernetic systems once you introduce this um so like you can have like and and if you have something like the big five showing up or the the big ten or the medium ten i don't know what you want to say but um they might show up differently so like you can think of like generalized agreeableness as like tendencies towards like non-zero um interaction and like not defecting in generalized game theory something like that like in a multi-agent context you're more of a cooperator and you you coordinate better but very different thing if you're actually having something like the feeling of compassion or empathy um these are different processes getting into the mix and you want a different handling and some things might not actually you want a very different handling yep interesting like would rather somebody act compassionately and be perceived as high percentile compassion or perceived overwhelming experience of compassion stephen marco colin yeah this brings in the idea of how people relate in a task-based environment or a

more task-orientated environment i suppose where it might structure things a bit more clearly between the things in the world where the person is and whatever mechanism or structure is interacting between the two i am i picking up at all am i there's some sort of delineation one thing that sort of comes to mind when we talk about so you've got the generative model you've got these hidden states of the world and you've got this markov blanket and the thing with the markov blanket is unlike it's not a system boundary you never can really define it so you've got this and what i thought was quite interesting really when by having the stability and plasticity i'm seeing sort of the the plasticity with this extraversion which you could say is like influencing the hidden states in a way you're perturbing out there and the um the intellect is making models is is the sort of the other side but the the the the um stability in some ways is really stabilizing with the markov blanket like the the nature of the regime because you've got of got you know neuroticism is it like a is it a very hidden landscape of actually like inferring so maybe we because i'm really interested in how we might have mental space psychology approaches which there might be some sort of intervening um spatial mediator to how we process and that spatial mediator of how big and scary things are around us and in our metaphors it could be sort of mediating so the neuroticism the agreeableness to conscientiousness seem to have more structuring about how i might take data in through a markov blanket to the other side and the other two seem to have a bit more about what's out there and what's in here in a in a very crude way but that's what i'm that's just something that came up for me i thought was really interesting cool thank you marco collin blue thanks um yeah i i also want to build up on that so i fully agree um so so i think i think the pro the um prospects that are most attractive with this kind of approach to personality is as stephen alluded to that that you can uh associate these aspect factors into particular imperatives to particular challenges as it were um and uh i think it's nice to know to connect that with what was mentioned in the beginning about mars levels of analysis so um mars levels of analysis is a bit of a hot topic always um so so forgive me i'm a bit biased but personally i don't believe that active influence should be seen as quintessentially computational because i think it would discredit the fact that it implies a lot about um the algorithmic level so um just like uh steven kind of already said you could have these intermediate spaces right and that's basically asking you how these computational imperatives or functions are going to be realized um and i really want to emphasize the notion of mediation because i want to connect that to what scott was talking about so um it should be emphasized that from what i can tell this model of personality um is kind of neutral about consciousness right it doesn't we should be very very um mindful

about not conflating the first person signs or perceptual models of how empathy arises within us with the process of empathy as such because if you see the pros of empathy within the brain then there's two ways to perceive it at the very least third person view from measures whether physiological or questionnaires and first person of course um and it stands to reason that that it's not necessary to have that first person encounter with these processes for this process to happen but having said that it would be very interesting to to to imagine that consciousness would be like a sp intermediate in space as well such that the component factors and the respective challenges of the big five um can actually equilibrate in the sense of that it's not a trivial challenge to balance the respective aspects of these factors and the factors between each other for the uh autonomous being of concern and so it would stand to reason that indeed um there might be another intermediate space intermediate between these five which would emphasize the cybernetic function of something like consciousness as a space which i think it might be interesting for you scott because i think then you can generalize it for non-living beings or or composite organizations because if you then associate personality as heuristics for certain domains of challenges or domains of imperatives and then uh having to bind them together through a certain integrative space then you could have i think a very interesting approach to organizational science if that makes sense thanks that reminds me a lot of the projective consciousness model paper that we read in a previous active stream but it's almost like wondering if the cybernetic organisms or cybernetic systems other than organisms if they have a projective geometry do they also have that awareness so colin blue scott what i wanted to say actually ties in nicely to the point that marco was just making i was going to start by saying that i think it's important that the existence of personality is not dependent on the existence of qualia so this is a i think a mistake that people often make when they're considering the big five and similar personality systems they think oh well these are just about how people perceive themselves or how about how people perceive other people and maybe they're useful categories in terms of like what you need to know about a person but actually the claim is that you know assuming that these descriptions of people's behavior have validity they describe patterns of behavior that exist regardless of whether people are accurately aware of them or not and so you could have differences in you know persistent differences in behavior from organism to organism in any species that would be directly analogous to what we think of as personality in human beings now of course it is true that we have this additional layer in which one uh kind of characteristic adaptation that we have is that we tell a story about ourselves consciously and we can describe ourselves and we actually make use of that

when we try to measure people's personalities because we you know often just ask them to describe themselves but we're not interested just in when in their description of themselves as such we're interested in or we hope and you know we know from a fair amount of evidence that those descriptions are reasonably accurate so it's not just that i think that i talk a lot i actually do talk a lot if you if you take some kind of objective measurement of that right that's part of what being extroverted is about um and so you know consciousness is obviously something that well is mysterious in various ways but i think that you know if we focus on that division between the easy and hard problems of consciousness i think we actually have some pretty important things to say about the so-called easy problems of consciousness like how does it function what does it do um and you know i think marco is right that one of the things that consciousness does is allow people to deal with conflicting information conflicting goals and to to create a sufficiently broad model of the world and our attention of it that we can then control our attention effectively i really like jeffrey gray's uh perspective on consciousness his last book was just called consciousness creeping up on the hard problem um that's really excellent and then uh i also like uh what is his name michael graziano's theory of consciousness it ties in very nicely to this idea that a cybernetic system uh to be a good cybernetic system you have to have some kind of physical representation of the variables that are being controlled that's the good regulator theorem um and so i think that one useful way to think about consciousness consciousnesses as a model of our attentional field that allows us to uh direct our attention effectively to things that are important for our goals or things that are potentially conflicting between goals and so in that sense yes it would certainly be relevant to conflicts that arise because of different needs or goals that are related to different personality traits but it's not something external to the traits mechanistically because the are descriptions of patterns of variation that you know in a population and so when we ascribe them to the person the person has a level of the trait and so those traits actually should be reasonably comprehensive descriptors of functioning in all the important you know categories of human function and that includes consciousness and in fact if you look at the contents of the big five openness and intellect is the trait that is really uh related to the qualities of consciousness people who are high in openness have are conscious of more things their is broader it's deeper it's more complex and you know so there's the openness side of that trait in the intellect side and so you've got this sort of pattern detection and then pattern analysis and all that is basically about uh in large part the qualities of our cool very interesting blue then scott then adam so i am going to totally bypass consciousness because it's just uh it's a hard problem

right but um i do want to to just consider for a minute what would happen if we maybe maps like not not asking what is a person but what is an individual could we then like scale over like non-human systems like you know organizations or even to other animals like adam brought up like major transitions in evolution and how do these like personality traits maybe influence evolution that would be interesting to think about like can we think of maybe clams as like ultra neurotic features and ants as like high in conscientiousness could we take like these personality traits and map them onto organisms you know that have evolved over time and and what would that look like in terms of of evolution and what what is essential you know openness to exploration is obviously key for you know some of the major transitions to happen that reminds me of the principal components the big fives um they're based upon the patterns of empirical variation in the actual population as colin brought up and then the animal spirits are without making a metaphysical claim they're like archetypes like of bravery or of extraversion or social butterfly or you know a couch potato it's still a living creature it's a potato it's in the ground and so how we model different types of metaphors it's like that's on our procrustean bed but then what does it look like to have it on their procrusty in bed for their personalities or individualities so scott then um marco then colin blue i love the notion of neurotic clams that's my favorite um yeah this is this is great and just along the same lines a couple of things that leaped into my head i was thinking of irving goffman stuff where you project yourself out as you want to be perceived so that's kind of you're modifying your environment in terms of the persona perception um that was one thing then also there's a lot of self other things in here the identity kind of notion and that and that idea of presentation of self one of the things i was wondering about is the dsm diagnostic statistical manual dsm-5 just came out and really how it plays into the um what i guess they would define as anomalous persona traits um and usually dsm seems to be the ones that aren't the ones that make you not productive in the capitalist society kind of things because there's a lot of things in there you call cultural in there but it's that those are the outer bounds of uh what's expected expectations of persona and i think a lot of the survivalness questions have to do with expectations and the uh the rising of these variables in the corollaries in animals and in people have to do with survival and increasingly social survival but also just that survival of the organism and so that idea of presentation of self uh might be even related to camouflage and issues of that you know how do you present yourself or not present yourself in society so anyway that just that host of thoughts that how that naturally evolved is kind of interesting also this is and this is not a a studied thing i'm not an expert on this but just functionally again i think of the um mind

and consciousness well i think put consciousness aside just like blue said but i think of the mind as not in the brain i think the mind as in language and material culture and the brain as an antenna tuned into the local mind so if you're my sister adopted a child at the age of five months and now she's a was raised in central pennsylvania and she was adopted from china she doesn't speak chinese she speaks spanish in english she's a u.s person but she looks like a chinese person so that their mind was developed in in in the in pennsylvania and so if there's no feral consciousness which let's take that as an assumption for a second with no feral mind and if it's a learned phenomenon it's interesting that we have this framing and then we keep re-upping the framing reaffirming and valorizing the framing it's interesting to wonder if there's a lag in terms of survivability what were the persona characteristics that were selected for and i don't just mean in biological evolution but socially are they identical to the ones that will be selected for in a much more dense interaction environment for instance so there may be some interesting questions if and to the extent that persona is not on board in the brain inherently but is a learned social phenomenon what is the how malleable is it and i mean i guess that's what education is about and why colonialists try to get indigenous people in their schools their children in schools early to render the externality innocuous but it'd be interesting from a perspective of current situations that we have in current policy challenges and public health challenges etc information challenges how might we handle those recognizing a more plastic uh persona so just a few thoughts cool interesting marco colin adam thanks um wait i forgot what i was gonna we'll come back okay well i think that there are certainly uh i mean certainly true that there are aspects of the mind and personality that are learned but then uh there are others that are evolved essentially and sufficiently genetically programmed that they would be present even in you know feral children but and that connection back to the idea of the evolution of personality uh when i when you know blue is talking about neurotic clams i was thinking about some of the work that's actually been done on the evolution of personality and animals some of it is very interesting there's a really nice paper from oh i know almost 15 years ago now by daniel nettle that talks about the evolution of personality and humans and other animals and so one of the things that he talks about are these experiments uh where you take uh a bunch of fish and you if you just naturalistically observe them you will see that there are differences in personality some of the fish are relatively bold some of them are relatively timid and cautious and what you can see is that if you systematically manipulate their environment you can actually change uh evolution of that population over time in interaction with the personalities of the individual and so the way that i think about this is that

personalities drive evolution but evolution also drives changes in personalities at the population level so what happens is that if you have a if you have those fish in an environment that has a lot of predators then the bold ones get eaten and the frequency of the more cautious timid ones increases in the population but if you put them in an environment that has very few predators then what happens is that the bold ones uh tend to monopolize a lot more resources and the frequency of the bold personalities increases in the population yes so always the ecological selection on the development of personality not even just personality something that can be reached out and grasped but actually how personality becomes developed in relation so adam then marco then stephen try to tie together a few threads before i lose him um so like earlier i was saying um you know evolution is one of multiple perspectives but a really special perspective and it is a really special perspective like i it almost makes me move towards like well on some level i'm uh an instrumentalist um but with respect to like the things like what is the ontological status of the markov blanket i think there is also like a realism realism and maybe a realism so strong we couldn't sometimes throw away with the word quasi we should always kind of keep it in our back pockets and that we should keep keep in mind that there are multiple ways of carving up the world actually infinite ways it's just some are going to be more or less good for being such as us and one really good carving would be the one that basically relates to functional cycles that achieve closure on time scales where they can be subject to cumulative evolution broadly construed whether being the development of the organism or phylogeny and so uh what that might uh pick out some particularly uh universally agreed upon while still subjective joints but to kind of bring it together more among those and with relation to your personality characteristics there's an interesting thing that can happen for species capable of modeling each other and themselves with a certain kind of richness having certain types of consciousness in the sense of knowledge where um and even without it potentially that they can become aware of these reliable course screenings of themselves and others and then this forms a basis for your planning and then this forms a basis for basically not just the niches that you're trying to construct but the way your niche is constructing you and so for all this like active inference with its like embrace an embracing of circular causality and this mutuality between like system and environment in terms of each being its own kind of active inferential system and exploring this interplay i think that's really essential for understanding personality in both the specific and the general case uh before i'll let you all move on like one final uh set of threads i would be like when talking about something like consciousness so like i recently kind of took on the hard problem uh it has

debated how much success i had but in trying to make way on this um it seems useful to not uh and probably with respect to discussions of personality um all of our words are just words they're uh course screenings that we agree upon um and they don't exist in the world and so consciousness part of the issue is like we mean very different things when we use the word sometimes and so um so similar so you might say um scott was suggesting like you know is we might want to distinguish between mind and consciousness or intelligence in consciousness and so you could say like your mind is distributed through the world and that your brain is an antenna i think that's a good perspective but if you're thinking of like consciousness as like a type of model where there's something that it's like from the inside to be that kind of modeling process that might not be extended that might be like a subsystem and a sub model um and so it's similar we're kind of building up this onion of complexity with respect to consciousness and people are kind of kind of engaging at different levels but not always being clear on the level that they're engaging it and so you could so recently um i was engaging at the level of like basic phenomenality itself um and uh rudroff and like projected consciousness models very similar spirit um it's uh you know specifically uh what are the ways we can describe your subjectivity as a particular reference frame and point of view in the world as you as a cybernetic system how can we characterize that and so then we go like another level still we're getting what colin's talking about with things like attention schemas and like models of your awareness for for modeling yourself and others and shared uh and shared attention schemas and so i guess the go further still into things like self-consciousness reflexive self-consciousness and each of these will be kinds of major transitions in evolution and development which i understand as a kind of evolution and each of these will now dock with personality in different ways um now i guess the one final thing with that's interesting uh is the way in which personality traits if you're thinking them as kind of coarse screenings it's like with like coarse grain descriptions and it's a one key core screen you can core screen when you have something like a renormalization process when you have some sort of like emergent attractor um such that the lower level or the smaller fluctuations uh smaller dynamics can be kind of like averaged out because now you're pointing at this bigger thing and the bigger slower attracting manifold didn't care about the particulars so uh rather that's not relevant for course or phenomena that you're trying to observe and so there's a sense in which um personality is a kind of correspondence like what are the nature of these like normalization processes and so one of it would seem like so what is inducing this attracting structure it seems like a good part of it would be so you can model yourself across situations so you know

what kind of agent am i that i am likely to be doing what and what circumstances and this will be pretty essential for pursuing any kind of complex plan but similarly for coordination um other working around other agents um or cooperation or competition what kind of agent is this i'm dealing with but for the sake of cooperation you might you know ask someone to be uh uh responsible they should be able to respond and report in a reliable and consistent way we actually demand of each other in a certain non-erratic character we we demand dimensionality reduction of each um just because otherwise we get surprised and as a friendly principal and active infrastructures we don't like being surprised um because then we can't plan and then we can't outsource our cybernetic entropy and it's it's bad so there's a sense in which um to what degree like there's this the same way that like personality traits are influencing they're emerging from evolution then influencing the course of evolution with development uh your personality traits are emerging from your interaction with your environment as you're carving out this niche um but also they can also become like attracting targets and of themselves on different levels like if you're uh if you have this expectation of yourself as being this type of agent it could be a self-fulfilling prophecy i think i'm a neurotic agent i think i'm an extrovert agent now other people think i'm neurotic agents other people think i'm an extroverted agent and now they're treating me a certain way and i'm acting a certain way and and then it keeps going um yes a lot of things you know a lot of no a lot of awesome stuff that reminds me of the scripts it reminds me of the phrase uh the common saying the absolute power to act absolutely brings responsibility or something like that like power and responsibility but now we're talking about the power to act and the responsibility to stimuli and that means that we respond so um then stephen thanks um adam for that fantastic um so first um a bit of a meta comment or question so since you said we can't use the chat in stream um i've been putting some links in the youtube chat but i'm wondering if you had a recommended way to do that because a lot of people are mentioning a lot of interesting research and i think people are modest to the point that they won't advertise their own research or link it yeah um so what would you prefer daniel sure to respond yeah to respond to the meta and just to clarify so in the jitsi for the live stream it will make a if you use the chat so it's very disruptive to use the chat during the jitsi so the single source of truth for the video is the youtube page or on keybase so definitely everybody should make a keybase username and join us at actinlab.public and check out activeactiveinference.org to find out more but the key base at this point is our actual organizational hub so that's where we're doing the open participatory active inference lab like this is a project of the communications branch of active continue okay

thank you yeah i didn't know but key basis but thank you glad i asked um yeah so building on saffron's uh great comments um so i fully agree that personality is definitely something scaffolded right um so um personally i i i i do see this at the the relevance of of course spreading but i think it's also um maybe interesting to use another uh example or metaphor of bootstrapping so um one interesting thing is so an ai keep dealing with interactability but clearly we are still living and not defying laws of computation so um there needs to be this kind of leak of faith of how to start off acting around in the world uh and so so one way to see it instead of terms of course grading in terms of temporal scale maybe it's simply um then put it this it's going to behave over time over time over time across different contexts there should be overlap in the kind of parameters or hyperparameters that govern your behavior approach to the world and those that are more consistent or more covariance or synergetic they will consolidate into this kind of heuristic core to further elaborate and specify um and on top of that building answer that i was wondering because adam earlier asked colin if you could maybe explicitly map the uh the big five to mechanisms cybernetic or inferential terms but i was wondering if it's um appreciated or useful to also put the question for everyone since everyone's relatively familiar with active inference or fustonia terminology how they would characterize um the big five factors because mostly it's in very simplistic uh adjectives or purely behavioral but i think would be interesting to maybe at least attempt faithfully to understand them from the inside out right what kind of functions or processes or inferential problems would be associated with them a few people already did that but i was wondering if people would like to uh do that more explicitly uh together cool great question and also a lot of it is enclosed in the paper which is why this is so fun to have as a journal club because we can always superfluously add on top but then the paper also references a lot of these things so we'll go to steven colin scott yeah a lot of really cool stuff um i think um well as well as the modeling idea i think going back to this the the the idea of a timid fish and a bald fish and the affordances of the environment one thing that's really been i've been really striking me as has been trying to integrate um drama-based projects and sort of experiential workshops with psychology is is that you've got this kind of reflective perspectival aspect to psychology like you can take a perspective and reflect and you get a sense of it the big five will use that in the reporting um yeah it's tapping into some sub-personal dynamics and other dynamics which may not be available at the conscious reflective level and i think that that that then starts to get really interesting with the affordances again with the timid and bold and what's been talked of of how this all sort of scales because um it's not necessarily a timid generative model that

creates the timing like if i take too many cups of coffee okay now my i don't know if it's my brain that's changed or whether it's the sensitivity to my um of my of my perceptual and active action reflexes which are kind of now or maybe my stomach or so you've got this um there's an interesting way with this that it can bring in the kind of the dynamics and the affordances and so one thing that's quite interesting in in this in theater you've got an acting method which is basically very like you hear a very text orientated and what did my character want to do but another way is you just actually get into the body of an animal so you be a spider and they actually have in the school they do like a set a whole series of weeks in the zoo and everyone has to be animals and inhabit animals bodies and but you know there's something different about if you're on four legs and you're moving around smelling things okay you're going to be balancing your inputs and your options and affordances for action are quite different right so it might well change the way um but it will change well i'm not going to speculate on that but just just basically coming back to the point of the timid and the bold and this idea that the the way the active inference or the inactive sort of dynamics of the body with the can be a lot of that could be sitting at a sub-personal level like you're not really sure how you know that bright light has affected my retinol or my tiredness has affected or my alcohol consumption you know some of these types of things so i thought i'd sort of say that that can connect in with this the more reflective piece and the sort of phenomenological cybernetic subconscious piece cool colin scott adam i just wanted to say that the reason that i didn't take up adam's suggestion to go through each of the big five and talk about the cybernetic processes that we think are probably related to them is precisely that they you know that is all discussed in the paper i didn't want to take up a bunch of extra time of things that um many people have already read um in terms of this kind of idea of the you know the richness of the psychological processes that are underlying each of the big five that is something you know it's true that a lot of people have this very shallow perspective on um partly due to the way that they were derived in the first place and those stereotypes that people have about them as you know scientific entities but i could spend the rest of the livestream talking about these processes in great detail but i think would probably be better just to refer people to the original paper that i wrote in 2015 on cybernetic big five theory and then also maybe to a theory that we wrote on the cybernetic perspective on psychopathology because the big five turn out to be equivalent also to the major dimensions of risk for different forms of psychopathology and we talked a lot about the specifics that might be important there as well and that all all my papers or most of my papers are available on my website cool great and that's also why again the journal club

format it's not the only format we're going to experiment with in active inference lab but this will actually allow us to have conversations that are rooted in a lot of specifics and various papers will be amenable to that in different ways so scott and then adam so and i wanted to apologize for not reading not having an opportunity to read the paper before this one so it would have been better informed had i done that one i wanted to um this just occurred to me something that might be going on here and it's a little bit stretched so let me give me a second here so this is going to be where i'm going with this is whether living forms are entropy engines okay that's where i'm going to go with this so um there was uh [Music] entropy i'm talking about now using shannon and von neumann and um entropy notions generally not by analogy but rather saying you're talking about general disorder and and so i'll be a little fast and loose with entropy for a second here so penrose wrote a paper a book recently we talked about the what we get from the sun is not energy but rather neg entropy because the yellow light of the sun comes in infrared light goes out the difference is the frequency the information carrying capacity of yellow light is higher because higher frequency and so it's a net energy neutral except for global warming or climate change but it's um what we get is the structure of plants reflects that higher neg entropy opportunity i guess is one way to put it so let's so starting with that um what if entropy was driving a lot more and so what if the um remember recall my definition of living forms as auto catalytic entropy secreting entities so if entropy universally is increasing living forms are localized neg entropy areas right they lower entropy locally and try to preserve it what if the the drive both the physical symbiogenesis where pre-existing bacteria migrate on board to eukaryotes and then become mitochondria in human in animals and chloroplasts in plants so they come together because it's a better deal basically a better way to be auto catalytic and entropy secreting at the larger scales what if that extends um to where we are now with the with the the what if those stage stages of perce personality have those functions we're saying they do and um then it has to do with both entropy accounting so when you're meeting someone who shares personality states or has predictable and ex personality states you can predict there's a lower um differential so from a second law of thermodynamics perspective you have like for carnot's equation you need a hot and cold differential to perform work what if the information differentials that you have between entities that are meeting each other help to cultivate various responses and those responses are we are cataloging in those five big persona traits and so that's ways of being or ways of encountering um so it helps you navigate the social landscape and um that the what what if what we have really is what i call an entropy engine which is that the personality goes out in

the and harvests disorder um through these mechanisms dissipating it obviously but also um growing from it and what i mean by that is all institutions all of them are uh dis de-risk some subset of interactions whether it's grocery stores or hospitals or banks or whatever that's what institutions do at larger scale so they're neg entropy expressions at larger scales but we have those um those those feelers that go out and maybe the personality types and our physical attributes are the feelers that go out to figure out where there's opportunities for proliferation into a community of interest around a shared risk thanks wow very interesting with the jordan peterson the darkness into light ordering the universe and you know or disordered order as a thermodynamic process so hopefully that will be something we can like continue thinking about in our last 20 minutes so we'll do adam then uh and anybody who's watching live can leave a comment anyone else who's on here can prepare any like last thoughts but this has been a super awesome discussion so adam then um with relation to entropy um you might want to check out the work of uh in terms of life jeremy england he has a really interesting handling of that's free energy principle compatible of entropy maximization and you might also enjoy terence deakins work colin hooked me up with him it's incomplete nature he gets into um discussions like cough uh auto cut discussions of auto catalysis like kaufman talks about and um uh work cycles as ways of like describing systems and their reality you know if you're talking about realities you know differences that make a difference uh being able to contribute to a work cycle is a pretty good criteria and if we're able to identify system on some sense it's engaging probably in some kind of work cycle and you can do an entrap like a handling of it that's going to involve entropy uh also in terms of entropy um there is maybe two papers you might like one is um i posted them in the youtube uh feed which is um uh psychological entropy um from uh hirsch mar and peterson 2012 and then a more recent attitudinal entropy framework um by daleigh bores boom harvard and van der masson forget my pronunciation uh but to now look back around to some of the things um that stefan was saying uh that still on some level abstraction will involve entropy um one of the places i'm actually really thinking uh now into and trying to understand is uh the conditions under which personality might be more or less plastic the conditions under which you can get trait change altered traits um and not just altered states and uh along these lines of notions of like acting like technical acting versus method acting like to what extent is this sort of self-fulfilling like the self-perpetuating attractor dynamic that makes up persons and their personalities um to what extent uh is it changeable even if there are these enduring cybernetic parameters to what extent can you take those arrows and columns causal diagram and really work them so collins

um basically talked about like if you want to have in his well-being work in his um uh psychopathology work um about character adaptations being where you have the most leverage like the things you tend to do the this like design of your life and the situations you show up in and how you show up and this is where you like or clinicians and yourself you might have the greatest purchase in an acting change um but and so this would work on multiple levels one would be like it would change the nature of what you experience and you know we're learners we update it would also change your niche construction and change the way that the the context in which you're in if you if you that would be leveraged there but um i guess to one more thought would be um and bring it back to um like to what extent it's it's not just chance so to change your characteristic adaptations would be to change the things you tend to do um but what you're describing this sort of like method acting view or are you bringing that up but like acting um what extent can personality be changed by really paying close attention to this what is it like to be me and actually and what could it be like to be a different kind of me and then imagine yourself in that way um i won't quite call it faking it until you make it but like um and to what degree is this a substantial part of how personality forms to begin with like children at play copying their parents pretending to be this fair that way but now the question would happen in the bootstrapping can we still leverage this um as adults and one thing along those lines would be that makes me think you might be able to um i believe in the uh tibetan tradition though um i think it's part of jokes i don't know how to pronounce it but basically um they'll do these kinds of like uh deity practices where like let's say you have some kind of like affliction something that's like not working for you um you might pretend to be a kind of complementary uh deity or spirit but you don't pull this off on your own like the social context and so you create this kind of sub persona that you live into i believe they call it tulpa marco i think actually pointed me to this um but uh but you don't do it on your own you have a holding in hardness social context that supports your act and you're able to kind of collaboratively enacted and that seems to me to be agree this is we can you know empirically establish this as a mechanism personality change this seems to be part of what makes the magic happen so uh yeah that's the things i'm sort of thinking into like what are all of the levers we have to actually design our personalities going forward so you know what mental techniques uh psychedelics like meditation what what's at our disposal so that we can have like you know we can good meta regulators over ourselves and realizing like our deepest values thank you adam i'm just gonna throw up that closing slide and then we'll hear from everybody who wants to raise their hand with the final note um this has really been an

awesome active stream thirteen point one and 2. so in the calendar event for those who are participating live there's a feedback form a survey which would be really appreciated and all of you are always welcome whether you're the author or not to participate in a future active stream you can go to the spreadsheet and find out the calendar but all of you would be extremely welcome to be in a group active stream like this again and then if anybody has an idea for an event that they want to arrange or like an interview or a follow-up discussion or a few minutes isn't enough during this conversation to talk about drama or consciousness or whatever it happens to be let's just plan that as an event so that we can always make sure to hold the space for hearing about everyone's perspective because it's just such a great stew when we can have it all going so uh marco and then stephen and then anyone else who wants to raise their hand on a last uh note thank you i agree this has been very very enjoyable so thank you danny um yeah scott thank you for the nice um metaphorical pictures so i fully agree that there's this beautiful notion of the feelers right but but i think it's also interesting to um first distinguish reflexive feelers and intentional or expressive feelers right so how do you navigate uh to the world and in that challenge how do you navigate yourself and so i think personality or yourself model regarding you as a person are more of those reflexive theaters and then subsequently you are more able to grip or control or regulate the way you navigate the worlds with your perceptual feelers more specifically the the recognition models by which we are able to take in the worlds because they are what are being compared with the sense data um and you also said something very nice about i i really like carnot engines as a picture i actually love that you mentioned that um and you mentioned information now information differentials uh is exactly what active inference is about right so so so the beauty is also there's a notion you mentioned entropy a lot but there's also a notion of relative entropy which is also known as the cool block library version which is exactly what underpins the entire formalism um because exactly like the carnot engine you want to minimize the temperature differential but in rain temperature is constant right similarly the energy consumption doesn't vary with more than order of magnitude and so the interesting interesting question here is okay to what extent is it like a carnot engine to what extent um does information just the minimization of information differentials play a part in this functional behavior so one very nice notion that i like which i got from a friend said burnett shout out um is coordinated equilibration so both in the case of normal conventional thermodynamic equilibration such as incarnate engines as well as in my opinion for active inference you're basically equilibrating the whole collection the complex covariance and co-conditional collection of information differentials and so no

matter what's by virtue of it being physically instantiated it needs to equilibrate physically which has some relation with informational equilibration so in other words one thing i'd like to say is basically we're letting nature do the work right we're already nature it has to equilibrate and it happens that that finding of equilibria is in itself homologous or analogous to particular algorithmic instantiations which if you're interested is also interesting in computer science because apparently there's some nice works that connect um ionic spin models at three models of magnets with a self-organized criticality right how the brain operates via avalanches or cascades of activity apparently these are formally implementable to solve basically most famous algorithmic challenges so i think hopefully you might find that interesting um right so yeah and so i'm not sure what your exact phrasing was but i want you to note that i think it's also a nice picture to imagine that what we're doing effectively oh yeah yeah you're talking about neg entropy so in fact what you're doing is you're you're partially deciding how you're going to be imprinted by the world right and that's the whole power of the whole predictive picture because the patterns that are out there in the world um are effectively the challenges themselves so when we internalize the patterns then we're more gripped or attuned to our or have a bigger affinity with the world so exactly right uh in that regard uh yeah sorry i forgot the rest of what you said um and regards for adam's beautiful point about spasticity of a personality um i also very much echoed kind of criticism slash question of the whole assumption that personality is static right it's rigid but i think what he said earlier about course grading and what i called bootstrapping kind of maybe is a path to understand better how it shouldn't be seen as so static because if it indeed is a bootstrap in a time where it lacks self models and basically the accessibility of changing that initial bootstrapping model is uh is is inhibited or it's obstructed effectively right because if you want to reflexively act upon these models then you need to have a path towards these models so that you have to um act upon whatever influence your personality through layer and layer layer from superficial to deeper right um and that's a big challenge and i think indeed things like topaz things like visualization exercises are one example but i also really like siemens notes on drama and more situated experiments situated design where i think the notion of porosities was very interesting right to what extent do you allow the environment to influence you um and finally i just wanted to note that's because i mentioned it uh psychedelics yeah absolutely because the whole phrase is psychedelic the revealing of the mind but that's only one step the second step is reveal such that it becomes accessible to to systematic change um so yeah so so those are my thoughts and thank you for giving me these thoughts nice it's really great how you tied so many of those

elements together so in the last couple of minutes just quickly from anyone who wants to speak we have stephen then scott then anyone else who raised than adam than anyone who raises their hand i mean this is what this paper is so interesting about is the way that it gives these this dynamic mechanism which is good for active inference but it's also quite an intuitive way to break down all these genetic models and and different things it gets hard when you start to layer different areas i suppose the two things that come up when i was thinking about the comments on the drama or acting is i think sometimes the acting or having a chance to go and do a performative event or action that's more like you're you're going on stage and that could be like that mounting car problem i don't know if you see if you see the paper of exploring a um using your model and seeing different possibilities so it could extend out the possibilities and that mounting car that problem was quite good for looking at how that issue of expanding out can be important and then there's also and there's different ways that for instance method action is actually can be quite dangerous a lot of people have psychological issues if they go down that route um so the result there's a basically putting on an act is like you're still a performer and you know that you're in a sort of a field of practice presenting and being aware of where the audience is how close they are what size they of you they can see however just by doing that you will have that experience so you will have expended your expanded your range your entropy of possibilities or states of the world there's also participatory type drama where you could revisit a real life experience and go back into um through psychodrama social drama an actual experience that you had and look at that and then that also could give different ways to maybe understand your actual lived life now whether that would change the big five i don't know it may not be about changing the big five for someone it may be something else within that you know but it um so i think it's really there's a i just thought i'd add that to the great competition okay thank you scott and then adam then colin just in the last couple of minutes this is great very rich discussion i like the fact that we raised more things and this is really the beginning of a discussion than we resolved and that's beautiful um the tilpa really struck me because it's something where there's an imagination in an internal which is re valorized reified by the external in a in a process way and it really reminded me of law in law if you say you have a right but there is no duty of anyone else to respect that right like if i say i have a right to free speech a right to sit anywhere i want on the bus and no one else respects it i don't really just i just have words on paper i really need the duties and the rights and that dyad i think is really interesting that reification that valorization and it started to wonder whether it's it feels like something like behavioral catalysts or

consciousness catalysts where you have that external and internal element are necessary for that coordinated equilibrium equilibration that as marco was alluding to so i'm fascinated by the discussion here i feel like there's we're just beginning and looking forward to other ways for us to continue because there's a lot of pieces that we still need to explore it feels like totally agreed uh so we'll have uh adam then colin and if anyone has thoughts on where we could explore next or what would be fun to explore they can mention it too adam then colin and a special thanks to both of you authors for coming for both of these conversations it was just it really made the whole thing come together so adam colin blue yeah i've been loving this thank you all of you um i'm really glad marco brought up criticality because it gives me an opportunity to hold forth for uh that's what colin knows is my religion the and i think this someone harkens back to last week's discussion of um meditation and mindfulness um so one of the ways um i view mindfulness is a kind of like psychological manifestation of so it's like there's this edge of chaos or this enter regime where you have enough uh variation enough disorder to give birth to a dancing star or rather to have variation for evolution to work with but then you have enough stability to get this cumulative evolution to have structure built upon and so on and so if it's the one regime where evolution is possible yeah this is a multi-scale phenomena this is probably this is a universality class like you'll find it everywhere at all scales and so at the psychological level um whatever you're really doing um in some sense i think it's going to be involved with respect to optimality and if one of the things you're doing is trying to uh for instance uh try out something new um and like regulate like the amount of psychological entropy you're experiencing um this um basically setting up for yourself this return to this metastable place where you can balance stability and plasticity where you can sort of ride along the edge of chaos and not fall in or or go too straight too far away from it um this is i think if you want to um either maintain your personality or change your or work within the bounds of it this seems to be really essential and uh i i this final thing is like i've been increasingly thinking like the way we think of mindfulness is like all wrong and that rather than thinking of it as like um this is a really nice state to be in and you're watching your breath and is like super great that's good but like more like the face space description and more as process and rather than like a thing you get it's a return to like a kind of a unnecessary home and really an essential part of cybernetic functioning of balancing stability and plasticity that you're going to need for anything thanks so much adam colin then blue i'm glad this whole idea of the edge of chaos came up because adam earlier you said at some point you know we're talking about active inference so talking about free energy and we and we don't like

surprise and of course as soon as i hear that you know what i think well no we do like surprise um uh we seek out surprise often we like surprises right we like uncertainty we like uh we like risk people enjoy these things that's exploration and so and the reason that we have both one i think maybe the most important fact in psychology that we have an innate ambivalent response to uncertainty and an increase in entropy which is simultaneously positive and negative right we obviously don't like surprise we fear entropy we need things to be predictable or everything disintegrates but at the same time we like surprise we like uncertainty we like exploration and if we don't do that then we don't adapt and eventually we get too rigid and everything disintegrates right so you have to seek out entropy in order to avoid eventual catastrophic increases in entropy right you have to seek it out and master it um and so that's why you need to be at line between order and and chaos right between order and and disorder and you know as adam's pointing out that's what uh that's what evolved systems evolve towards is toward the edge of chaos because that maximizes the possibility for computation it maximizes the possibility for adaptation um and there is to make a segway from the abstract to the more creep because i feel like that's sort of like what my collaboration with adam it's like i'm tying this to more concrete psycho psychology he's trying to more abstract theory there's a lot of work in the last 20 years in personality psychology on personality change and especially personality trait change one of the things we know from this research is that virtually everyone has at least one personality trait that they would like to change we also know that personality traits can and do change over time it might not always be easy to change traits they might not always change you know dramatically but we do see change in personality traits over time we know that things like therapeutic interventions change personality traits especially reducing neuroticism and personality psychologists have recently been trying to develop uh systematic programmatic interventions to help people to change different personality traits and of course a lot of it is through first just changing more specific habits more specific characteristic adaptations because that's what as adam said you have more of an immediate grasp on is how to uh how to change your habits how to change your particular strategies but it is possible for those to generalize to these more uh general changes and i would think of that as you know you you learn specific things but eventually those things that you learn may actually tweak some of the underlying general parameters that actually enact those specific things thank you colin for those who need to drop off thanks for coming just to respect the end of the official event but blue dave and then we'll chill for a couple more minutes but thanks everyone so much for coming today um so i want to thank the

authors especially for for being here and livening up this conversation but just as far as things that i think we could continue talking about or that would be interesting to pursue in the active inference framework um are things like personality change or things that do change the personality right like trauma therapy psychedelics mindfulness meditation like all of these things can really have drastic effects on who a person is um fundamentally i i expect in like terms of big five as well and then also like maybe in a non-active inference kind of way like the discussion on consciousness is is big and heavy and uh maybe not for this live stream but something that is definitely always worth um talking about thank you blue dave yeah um sean carroll interviewed carl frist in a few months ago and he brought up i think the same point professor de young just mentioned wait a minute we don't try to be certain about everything we want novelty we want we want to be confused and professor christian became even slower and more methodical and self-critical and self-qualifying than usual and really went i think was working through all these levels of mathematization that i'm sorry in my humble opinion he's kind of gotten trapped into that's something that professor edelman was terrified as this incredibly brilliant young kid and he's gonna get his brain sucked by the algebra thank you dave um you know it reminds me it's controlled novelty we can't just say more it's more as different as what we say in complexity when it comes to cybernetic goal-driven edu chaos systems it's not so simple as just take a linear aggression and you know more ants is better for the colony no i mean selection has clearly in different niches resulted in colonies that have different shapes of everything so it's a local vector of selection but then the broader space is actually not quite so linear so i hear you on that yeah thanks i'm glad you mentioned that podcast actually uh in my opinion the best conversation with carl because sean being also professor carol also being a physicist he really was able to get carl um active and and it's more playful and cheerful about talking about it but i'm not sure what you mean with the with the getting more required and stuff critical i think it's more is awareness that that's really the the main challenge in dealing with these kind of seeming tensions or or paradoxes because i do think there's a very important intention to point out when it comes to formal modeling especially when it comes to formally modeling brains or humans as we are all also familiar with um which concerns um as seen for example in the personality model and there are often these complementarities these trade-offs um and often in infor psychology in normal communication we tend to pick a trade or pick a characteristic and just say that's good or bad it's always decontextualized it's rarely in relation to the trade-offs um in relation to which they obtain an actual normative value or valid normally value might be um and crap something else

oh yeah so um i link to the youtube on the youtube chat but but i really personally am very fond of the theme of pleasant surprises pleasant perturbations um and this controls on these controlled unknowns for example um well steven's gone now but but he mentioned the idea of space right so we could imagine um well first should be more formal so formally um that's already resolved in 2017 um a paper i was happy to be the co-author on um because if active influence doesn't concern itself with free energy per se it's expected free energy right and the more sophisticated your models the more sophisticated your temporal death and your contrafactual imagination uh imagination is very formal but i but i like allowing us to talk about imagination as if it was formal right kind of liberating it's from psychological connotations and way too many centuries of philosophy kind of dunking on it right because it's simply because you didn't have the grounding so so it takes exactly an active influence that these tensions disappear if we allow ourselves to to to accept uh what the mathematics actually say but there is this kind of trade-off but there is this kind of complementarity that indeed it is kind of the hidden notion of yin and yang um and so it's no surprise that stability and plasticity is the meta trade in my opinion um and oh yeah sorry and i linked uh that paper from heinleisch who is uh at the for university of brussels and a pretty prominent research ceremony and he has a very interesting notion which quite like about um challenge propagation so um that's continuing the theme of pleasant surprises because we design challenges for each other social situations games work all these cultural niches our designs challenge is simply by virtue of the fact that they were selected by our social cultural dynamics we allowed these to persist throughout time in culture knowing that in their persistence they enact positive effects on us as we try to adapt to them right and so i in that sense i find it very very beautiful um that these kind of controlled unknowns is controlled chaos is such a common theme in how we design each other and allow each other to cultivate and in that sense to another theme we co-regulate each other and in that co-regulation of each other we develop these kind of super personal personalities which i think well hopefully people can agree that we sometimes really feel that viscerally the kind of personality or vibe as as millionaires us like to say and there's some certain vibes in aries wow and these vibes tend to resonate with our personalities because we already have these models of this kind of strategies and goals and values we we instantiate for particular situations and so it's not far-fetched that the same models afford us also send certain personality-like themes in situations because that is literally the kind of brain system we use we use the brain system of persons to do this right the the brain system somewhere frontal there's a lot of performance area um we have next to each other a brainery activated when we

do self-modeling and next to that modeling other people and so that is also exhibited in how easily we attribute personality or intention to inanimate super simplified objects right and that speaks to in my opinion the validity of the core assumptions in paper uh that personality can be decomposed and indeed that's been perfected of values strategies and goals um sorry i went on a ramble i thought i was going to say but but yeah our relational architecture the way that we talk about organic chemistry and this chlorine wants to do that and this galaxy exactly that the intention the intentional stance that relational thinking is so deep and the way that you put it with a super personality it's like the superman is the collective leviathan it's not just the better individual nest mate it's that and in the context of something that's actually super or supra like above not just better um scott and then anyone else who wants to make a final comment but this has been really great marco this is i mean that was so beautiful he's starting to think of the choreography of consciousness you know that's what it felt like you know it's such a beautiful notion i always used to tell my kids now they're in the 20s i used to say the only time you see reality is when you see paradox and any other time you're inside a model and the paradox and that goes right the paradox is the reality that's the reality of our interaction you know you're interacting you know you know you're in encountering something and so that notion of embracing the paradoxical and the dance and the and the negotiation of it it reminds me the last point i was going to make is it reminds me of the research they did on fish swimming they took a semi-rigid piece of plastic and they put it in a stream and they started increasing the speed of the stream and at a certain speed of the stream the plastic undulated with a periodicity that was akin to how fish swim and they realize that fish don't swim through water they pass through the water and because they the water is causing their swim similarly that choreography of consciousness is that right we're not we don't just situate our cognition we situate our persona and our and and it's constructed externally and internally so the last thing i wondered about is is each system gdelian so each system is incomplete has questions within it that can't be answered from within it and maybe our novelty seeking is an intrinsic recognition that we're all gdelians that we need to go outside to find answers so anyway great discussion love it thank you guys thanks as always coming on scott um so uh let's just go from uh dave colin marco and then we'll probably call it unless someone else has a burning desire the only time we need to be conscious is where we're in that zone of paradox we ought to understand it and we don't nice colin well on the same theme i really like that idea of paradox being reality too and that's the same as the prima materia of the alchemists right it's what is the thing behind that gives birth to those things and it's precisely the thing that we don't

understand it's the thing that's unknown right and so that's the chaos that's on the you know the other side of the border where we're at the edge of chaos uh and yeah from especially from the perspective of consciousness what you need to be conscious of is precisely only those things that you don't understand perfectly already yeah um uh yeah scott thank you that was um uh yeah i just wanted to riff on that so so i i laughed because um i mean i feel comfortable saying this here so so you know me and my friends we were interested in psychedelics so so i find it very very fascinating that there's a huge variety in what i call idiosyncratic encounters so a lot of my friends with the psychedelics have this one thing that they feel like oh wow that was so fundamental and one of my friends uh let's call her luta um she had exactly that right she said i couldn't escape the idea that everything is parallax right and uh though i won't go into it but every other such quintessential encounter it is a great culture also in my opinion relates to something formalizable in terms of active um and i really really like the notion of choreography of dance i i use the metaphor a lot too the um you might also be interested in fridge of kaplan's who also wrote the tao of physics and he also had this moment of insight about physics simply being this uh dance right he used a metaphor of god a deity in hinduism but more formally and interestingly well my attempt to form formally make it interesting is as we noted right cybernetics acts upon configurations of models simply because systems are composite or compositional and distributed and constantly reorganizing so the very least can say that it's about configurations yes um going back to an earlier thread if conscious indeed is intermediating cybernetics then they need to intermediate the balancing or the meta configuration of configuration which themselves are constantly changing so exactly in in the example you took with the fish they are embedded or immersed in fluid dynamics right there are flows and i say that formally these are informational flows so they're immersed in all these flows and they need to find or the space needs to find uh the space need to situate itself in between those flows and by virtue of doing that it is regulating them right and so i think that's a very beautiful idea um and the last thing i think is also very beautiful so um we're all good earlier we're all in completes uh which reminds me of conversations i've had with a simon de deo a research at sfi institute very very good researcher very interesting has some very very nice work on bayesianism and the nature of knowledge and and um i i think i said to him exactly what you said right they saw kadalien but but how i phrase it is we're all each other's oracles so an oracle and you know in computer science but they give you an answer that you can't actually compute but if you would have the answer then things would come easier so it is by virtue of the noise we are incomplete and that is allowed because other people's incomplete

systems can act as an oracle for us and so i think that's also something that speaks to our experience of how we aware certain things but we're it's in in fact actually very complete but we trust that it's okay to be complete and if interested in that then uh i'm sure you know hofstadter goddess terence deacon in complete nature as adam mentioned and the day as i mentioned so yeah thanks so much that was very simulating great thank you a bit interesting i closed it with william blake quotation he wrote in his notebook do what you will this life's a fiction and is made up of contradiction they are beautiful i applaud to that peace out everyone thanks for participating thanks for watching it hope to like have more participation join the stream if you're not on it already and we'll see you next time thank you so much for watching this bye thanks everyone